

SUB-PROJECT MANAGEMENT PLAN (SPMP)

Analytics & Community Insights Module

Verified Neighborhood Community Hub

National University of Computer & Emerging Sciences (NUCES)

Version History

Version	Approved By	Rivision Date	Description of Change	Authors
1.0		30 March 2026	Initial SPMP – Assignment 2 submission, aligned with Project Charter	Shace tariq, shah faisal , hamad khan, ali sajjad .

1. Executive Summary

The Verified Neighborhood Community Hub – is a community-based digital platform connecting residents within their neighborhoods. Authorized on 12 March 2026, the project aims to replace fragmented informal channels (such as WhatsApp groups) with a structured, verified, and modular platform for local news, safety alerts, local services, events, marketplace, and community management. The project is managed by the Integration Group (Hassan Qureshi, Ahmed Ali, Muhammad Irfan, and Muhammad Hamza) and developed collaboratively by thirteen module teams across Section A.

The Analytics & Community Insights module is one of thirteen sub-projects within Bring. It is responsible for collecting, processing, and visualizing behavioral and engagement data generated across all platform modules. It provides real-time and historical dashboards for administrators and community moderators to monitor neighborhood health, engagement patterns, safety incident trends, and platform-wide usage metrics.

The entire project uses the MERN stack (MongoDB, Express.js, React.js, Node.js) under a zero-budget constraint — no paid APIs, services, or premium tools are permitted. Free-tier cloud platforms (Railway, Render, Vercel, Netlify, MongoDB Atlas) are used exclusively. The project is projected to conclude by the end of April 2026, with Assignment 2 documentation submitted by 30 March 2026.

Key success criteria for this module include accurate event ingestion from all other modules, fast and reliable analytics API responses, a functional admin-facing dashboard, and seamless integration into the Bring system during the final integration sprint.

2. Project Management Approach and Governance

The Analytics & Community Insights module follows the governance structure established in the Section A Project Charter. The Integration Group — comprising Hassan Qureshi (i233029), Ahmed Ali (i233054), Muhammad Irfan (i233065), and Muhammad Hamza (i233032) — holds overarching authority over interface definitions, integration deadlines, technology compliance, and architectural decisions. Module teams operate autonomously within these boundaries.

Management Approach

An Agile-influenced iterative approach is adopted for this sub-project, adapted for a group context. Work is organized into two-week development sprints aligned with project milestones. Progress is reported to the Integration Group via whatsapp (the designated project communication tool) weekly and at each integration checkpoint.

Technology Stack

- Frontend: React.js (MERN stack – mandatory per Project Charter)
- Backend: Node.js with Express.js (MERN stack)
- Database: MongoDB (MERN stack – free Atlas tier)
- Deployment: Railway or Render (backend), Vercel or Netlify (frontend)
- Version Control: GitHub (free student account)
- API Testing: Postman (free tier)
- No paid tools, APIs, or premium services permitted

Communication Channels (per Project Charter)

- Primary tool: ClickUp — used for task assignment, progress tracking, and inter-team communication
- Separate ClickUp channels for Team Leads and individual module teams
- Integration coordination: direct contact with Integration Group leads via ClickUp and WhatsApp

Roles and Responsibilities

Name / Role	Title	Responsibility
Shah faisal (23i-0058)	Team lead	overall coordination, communication with Integration Group, ensuring deadlines and deliverables are met
Shace tariq (23i-3069)	Backend Developer	development of APIs, data processing, database design, and ensuring backend integration
Ali sajjad (23i-0110)	Frontend Developer	development of dashboard interface, data visualization components, and UI integration
Hamad khan (23i-3095)	QA & Testing Engineer	QA & Testing Engineer – testing, validation, bug tracking, ensuring system quality and reliability

Assumptions and Constraints

- Project authorized and started: 12 March 2026.
- Assignment 2 documentation deadline: 30March 2026 at 11:59 PM.
- Development must be integration-ready by end of April 2026.
- Zero budget: no paid tools, APIs, or cloud services — free tiers only.
- MERN stack (MongoDB, Express, React, Node.js) is the mandatory technology stack per the Project Charter.
- The Analytics module must expose REST APIs conforming to contracts approved by the Integration Group.
- All modules share User IDs and Neighborhood IDs as common identifiers for data correlation.
- Continuous coordination with other module teams is required for event data ingestion.

2.1 Project Scope

Scope Statement

The Analytics & Community Insights module collects, aggregates, and presents data-driven insights from all active modules of the Bring platform. It enables platform administrators and community moderators to understand user engagement, neighborhood activity levels, incident frequency, marketplace transactions, service request trends, and moderation effectiveness. The module serves the project objective of delivering analytics and insights for admins and moderators as stated in the Project Charter.

In Scope

- Event data ingestion from all Bring modules via REST API using shared User IDs and Neighborhood IDs.
- Storage of aggregated analytics data in a dedicated MongoDB collection.
- Backend Node.js/Express API exposing analytics endpoints (engagement stats, post activity, safety trends, user growth, module-level KPIs).
- Admin-facing React dashboard showing charts, graphs, and summary statistics.
- Time-series reports: daily, weekly, and monthly views.
- Community health score computation based on post activity, incident reports, and moderation outcomes.
- Exportable reports (CSV) for administrators.
- Anomaly detection alerts (spike in incidents, spam, etc.) communicated to Integration Group API.

Out of Scope

- Real-time predictive AI/ML modeling (deferred to future version).
- Individual user-level surveillance or privacy-invasive profiling.
- Data collection from external sources outside the Bring platform.
- Automated enforcement actions (handled by Moderation & Conflict Resolution module).
- Financial reporting or payment analytics (handled by Community Marketplace module).
- Phishing or spam detection logic (handled by Phishing/Spam/Fraud Detection module).

2.2 Deliverables

Process Deliverables (Documentation)

#	Deliverable	Description	Target Date
1	SPMP (this document)	Sub-Project Management Plan covering scope, schedule, WBS, stakeholders	30 Mar 2026
2	SP-RMP	Requirements Management Plan for the Analytics module	30 Mar 2026
3	SP-SRS	Software Requirements Specification for the Analytics module	30Mar 2026
4	SP-SDS	Software Design Specification (architecture, DB schema, API design)	30Mar 2026

Product Deliverables (Software)

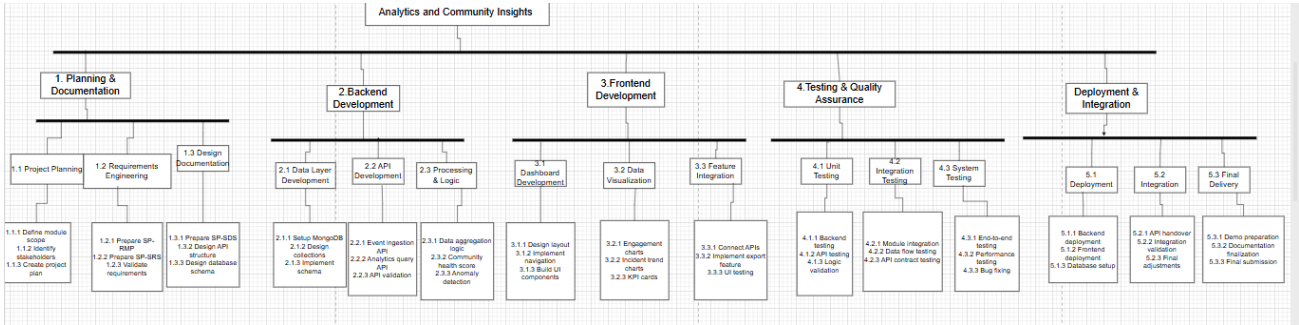
#	Deliverable	Description	Target Date
5	Event Ingestion Service	Node.js/Express microservice accepting events from other MERN modules	5 Apr 2026
6	Analytics API	REST API exposing aggregated data endpoints to admin dashboard and integration layer	12 Apr 2026
7	Analytics Dashboard (React UI)	React frontend dashboard with charts and summary views for admins and moderators	18 Apr 2026
8	Integrated & Tested Module	Module fully integrated with the Bring system and passing system integration tests	End Apr 2026

2.3 Work Breakdown Structure (WBS)

The WBS decomposes the Analytics & Community Insights module into manageable work packages aligned with the MERN stack. Level 1 = module; Level 2 = phases; Level 3 = work packages.

WBS ID	Work Package	Description
1.0	Analytics & Community Insights	Top-level module – delivers insights for admins and moderators
1.1	Planning & Documentation	All project management and documentation activities (Assignment 2)
1.1.1	SPMP Preparation	Prepare Sub-Project Management Plan
1.1.2	SP-RMP Preparation	Define requirements management approach
1.1.3	SP-SRS Preparation	Document all functional and non-functional requirements
1.1.4	SP-SDS Preparation	Design architecture, MongoDB schema, and API contracts
1.2	Backend Development (Node.js / Express)	All server-side MERN stack development activities
1.2.1	MongoDB Schema Design	Design analytics collections: events, aggregates, neighborhood stats, reports
1.2.2	Event Ingestion API	Implement POST /api/events endpoint to receive events from all modules; validates shared User ID and Neighborhood ID
1.2.3	Data Aggregation Service	Scheduled Node.js jobs to compute daily/weekly engagement and activity aggregates
1.2.4	Analytics Query API	Express REST endpoints for dashboard to fetch aggregated data per neighborhood
1.2.5	Community Health Score	Algorithm to compute and expose neighborhood health score from post, incident, and moderation data
1.2.6	Anomaly Alert Service	Detect spikes in incidents or spam events; notify Integration Group API
1.3	Frontend Development (React.js)	Admin dashboard React UI development
1.3.1	Dashboard Layout & Navigation	React admin dashboard shell, sidebar navigation, and header
1.3.2	Engagement Charts	Charts: daily active users, posts per day, new registrations per neighborhood
1.3.3	Safety & Incident Trends	Charts: incident types, frequency, and resolution rates by neighborhood

1.3.4	Module-Level KPI Cards	Summary stat cards per module (marketplace, services, posts, events, messaging)
1.3.5	Report Export (CSV)	CSV export functionality for selected date ranges
1.4	Testing & Quality Assurance	All testing activities
1.4.1	Unit Testing	Test aggregation logic and API endpoints individually using Jest or Mocha
1.4.2	Integration Testing (Module-level)	Test event ingestion from mock module events; verify all API responses and MongoDB queries
1.4.3	System Integration Testing	Participate in cross-module integration testing led by Integration Group; validate shared User ID and Neighborhood ID flow
1.5	Deployment & Handover	Final deployment and demonstration preparation
1.5.1	Deployment to Free-Tier Cloud	Deploy Node.js backend to Railway/Render; React frontend to Vercel/Netlify; MongoDB Atlas free tier
1.5.2	API Handover to Integration Group	Provide API documentation and live endpoint URLs to Integration Group via ClickUp
1.5.3	Demo Preparation	Prepare module demo script and participate in final Bring system demonstration



2.4 Stakeholder Analysis

Stakeholders are identified from the Project Charter's Roles and Responsibilities table and from module dependencies within the Bring system.

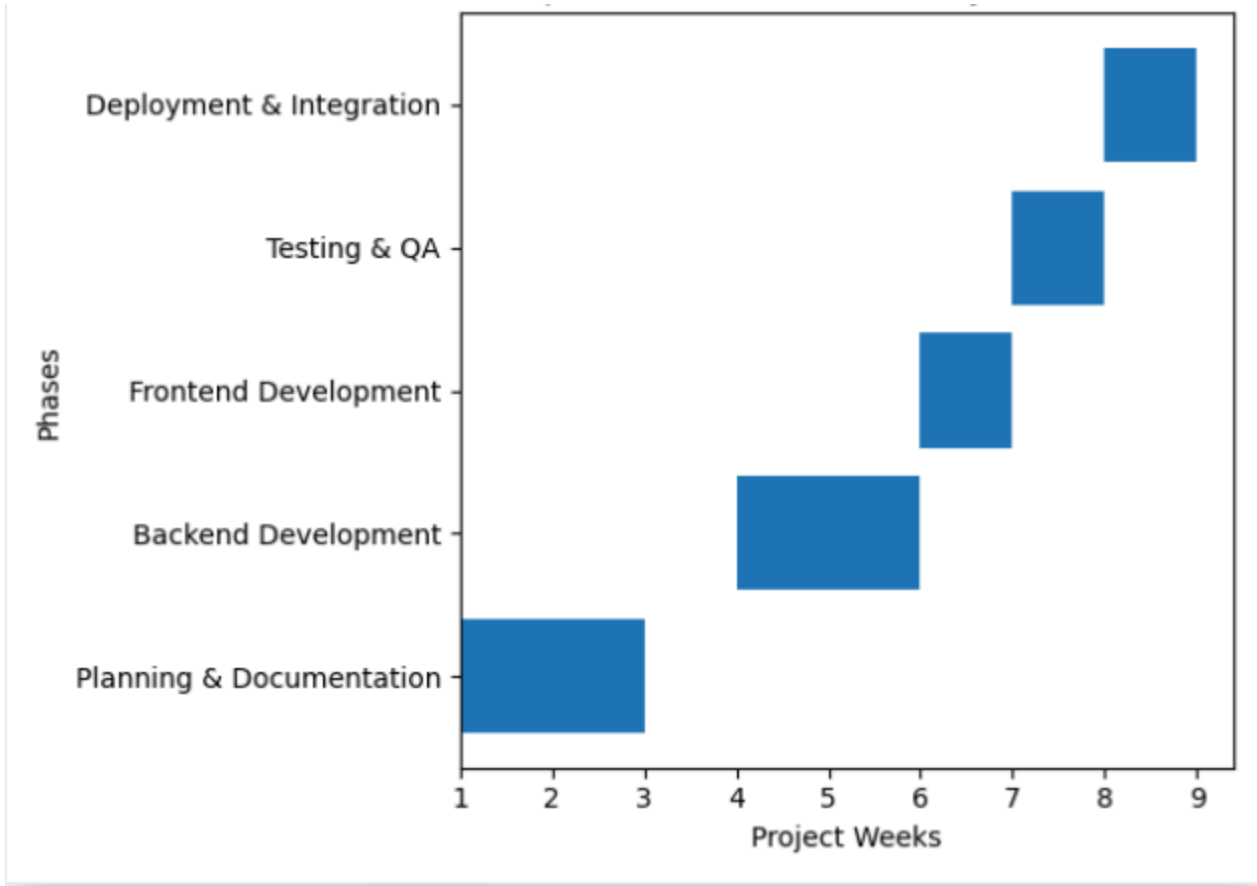
Stakeholder	Role	Interest / Impact	Influence	Strategy
Hassan Qureshi / Ahmed Ali / M. Irfan (Integration Group)	Integrators	Defines interfaces; depends on Analytics API for overall integration; manages ClickUp	High	Weekly updates via ClickUp
Rabeeha Kamran (Auth Module)	Data Provider	User authentication events (login, registration) must be sent to Analytics ingestion API	Medium	Share event schema early
Areej Hamid (Neighborhood Verification)	Data Provider	Neighborhood verification events feed neighborhood health score computation	Medium	Align on Neighborhood ID format
Maryam Iftikhar (Community Posts)	Data Provider	Post creation and interaction events are core engagement metrics	Medium	Event schema alignment
Aqsa (Moderation / Conflict Resolution)	Data Provider	Moderation actions feed conflict resolution analytics and health score	Medium	Coordinate on moderation event types
Shah Faisal (TBD Module)	Data Provider	Events from assigned module feed module-level KPI cards in dashboard	Low	Inform on event schema
Platform Administrators	End Users	Primary consumers of analytics dashboards for neighborhood management	Medium	Design for usability
Course Instructor	Evaluator	Grades the sub-project; evaluates documentation quality and system functionality	High	Meet all rubric criteria

2.5 Schedule Baseline

The schedule is aligned with Project Charter milestones: kickoff completion by 18 March, Assignment 2 documents by 27 March, development through April, and final integration and submission by end of April 2026.

WBS	Activity	Start	End	Days
1.1	Planning & Documentation	12 Mar 2026	27 Mar 2026	16
1.1.1	SPMP Preparation	12 Mar 2026	27 Mar 2026	16
1.1.2	SP-RMP Preparation	18 Mar 2026	27 Mar 2026	10
1.1.3	SP-SRS Preparation	20 Mar 2026	27 Mar 2026	8
1.1.4	SP-SDS Preparation	22 Mar 2026	27 Mar 2026	6
1.2	Backend Development (Node.js / Express / MongoDB)	28 Mar 2026	12 Apr 2026	16
1.2.1	MongoDB Schema Design	28 Mar 2026	30 Mar 2026	3
1.2.2	Event Ingestion API (POST /api/events)	31 Mar 2026	3 Apr 2026	4
1.2.3	Data Aggregation Service (scheduled jobs)	4 Apr 2026	7 Apr 2026	4
1.2.4	Analytics Query API (GET endpoints)	7 Apr 2026	10 Apr 2026	4
1.2.5	Community Health Score Algorithm	10 Apr 2026	11 Apr 2026	2
1.2.6	Anomaly Alert Service	11 Apr 2026	12 Apr 2026	2
1.3	Frontend Development (React.js)	10 Apr 2026	18 Apr 2026	9
1.3.1	Dashboard Layout & Navigation	10 Apr 2026	12 Apr 2026	3
1.3.2	Engagement Charts (Chart.js / Recharts)	12 Apr 2026	14 Apr 2026	3
1.3.3	Safety & Incident Trend Charts	14 Apr 2026	16 Apr 2026	3
1.3.4	KPI Cards & CSV Export	16 Apr 2026	18 Apr 2026	3
1.4	Testing & Quality Assurance	13 Apr 2026	22 Apr 2026	10
1.4.1	Unit Testing (Jest / Mocha)	13 Apr 2026	16 Apr 2026	4
1.4.2	Module Integration Testing	17 Apr 2026	19 Apr 2026	3
1.4.3	System Integration Testing (cross-module)	20 Apr 2026	22 Apr 2026	3
1.5	Deployment & Handover	21 Apr 2026	End Apr 2026	5+
1.5.1	Deploy to Railway/Render + Vercel/Netlify + MongoDB Atlas	21 Apr 2026	22 Apr 2026	2

1.5.2	API Handover via ClickUp to Integration Group	22 Apr 2026	23 Apr 2026	2
1.5.3	Final System Demo Preparation & Participation	23 Apr 2026	End Apr 2026	3+



2.6 Milestone List

Milestones are derived directly from the Project Charter schedule milestones and module-level checkpoints.

#	Milestone (aligned with Project Charter)	Target Date	Status
M1	Project Authorization – Project Charter signed; project started	12 Mar 2026	Complete
M2	Module Assignment & Kickoff Completion – module assigned; kickoff meeting attended; ClickUp setup	18 Mar 2026	Complete
M3	Assignment 2 Submission – SPMP, SP-RMP, SP-SRS, SP-SDS submitted via portal	27 Mar 2026	In Progress
M4	Event Ingestion API Live – POST /api/events deployed and API contract shared with Integration Group on ClickUp	5 Apr 2026	Planned
M5	Full Analytics Backend Operational – all Express API endpoints live and tested with Postman	12 Apr 2026	Planned
M6	React Dashboard Complete – admin dashboard connected to live MongoDB backend and rendering correctly	18 Apr 2026	Planned

M7	Integration Preparation – module approved by Integration Group; passes cross-module integration tests	22 Apr 2026	Planned
M8	Final Integration & Submission – module deployed, integrated into Bring system, and demonstrated	End Apr 2026	Planned

2.7 Change Management Plan

All changes to the Analytics module that affect integration points must be routed through the Integration Group and tracked on ClickUp.

Change Request Process

- Changes to API contracts or MongoDB schema affecting shared User IDs or Neighborhood IDs must be approved by the Integration Group before implementation.
- Internal changes (refactoring, performance improvements) within the module may be made autonomously.
- All approved changes are reflected in updated SPMP documentation and communicated on ClickUp in the module's team channel.
- API contract changes after 31 March 2026 are subject to Integration Group veto.

Change Log

#	Date	Change Description	Requested By	Status
–	–	No changes recorded at time of initial submission	–	–

2.8 Project Scope Management Plan

Scope Definition Authority

The module team holds authority to define internal scope. The Integration Group has veto power on any scope element that affects shared interfaces or API contracts per the Project Charter.

Scope Measurement

- Scope adherence is measured by tracking completed WBS work packages on ClickUp against the baseline WBS in Section 2.3.
- At each weekly check-in, a completion percentage is reported per WBS phase on the designated ClickUp channel.
- Any deviation affecting the API surface is escalated to the Integration Group within 24 hours via ClickUp.

Scope Acceptance

- Product deliverables are accepted by the Integration Group upon demonstration of working API endpoints and a functional React dashboard.
- Documentation deliverables are accepted upon submission via the course LMS portal by 27 March 2026.
- Final scope acceptance occurs during system integration testing in the final week of April.

3. Communication Management Plan

Communication follows the structure defined in the Project Charter. ClickUp is the primary project management and communication tool for Section A.

Communication Type	Audience	Content	Frequency	Channel
Weekly Status Update	Integration Group	Progress on WBS items, blockers, upcoming tasks	Weekly	ClickUp (module channel)
API Contract Notification	All Module Teams + Integration	Event schema or analytics endpoint changes	As needed	ClickUp / Team Leads channel
Integration Checkpoint	Integration Group	Live API demo, milestone confirmation	Each milestone	In-person / ClickUp
Document Submission	Course Instructor	SPMP, SP-RMP, SP-SRS, SP-SDS	27 Mar 2026	Course LMS Portal
Final Demo	All Stakeholders	Live demo of Analytics module in integrated Bring system	End Apr 2026	In-person

Project Staff Communication Contacts

Name	Role	Email / Contact
Shah faisal (23i-0058)	Team lead	i230058@isb.nu.edu.pk
Shace tariq (23i-3069)	Backend Developer	i233069@isb.nu.edu.pk
Ali sajjad (23i-0110)	Frontend Developer	i230110@isb.nu.edu.pk
Hamad khan (23i-3095)	QA & Testing Engineer	i233095@isb.nu.edu.pk

4. Resource Management Plan

All resources are free-tier and open-source per the zero-budget constraint in the Project Charter. No procurement is required.

Resource	Type	Justification / Free Alternative
MongoDB Atlas	Database (MERN)	Free 512MB cluster – mandatory MERN stack database
Express.js	Backend framework (MERN)	Free, open-source – mandatory MERN stack
React.js	Frontend framework (MERN)	Free, open-source – mandatory MERN stack
Node.js	Runtime (MERN)	Free, open-source – mandatory MERN stack
Railway or Render	Backend cloud hosting	Free tier – zero budget compliant
Vercel or Netlify	Frontend hosting	Free tier – zero budget compliant
GitHub	Version control	Free student account – shared with Integration Group
ClickUp	Project tracking / Communication	Free tier – designated primary tool per Project Charter
Postman	API testing	Free tier
Chart.js / Recharts	React charting library	Free, open-source – for React dashboard charts

5. Human Resources Management Plan

This sub-project is executed by a team of four members, each assigned specific responsibilities derived from the Work Breakdown Structure. Tasks are distributed across backend development, frontend development, testing, and project coordination to ensure parallel progress and timely delivery.

5.1 Project Staff List

Name	Roll No.	Role	Assigned WBS Tasks
Shah Faisal	23I-0058	Team Lead	1.1 Planning, 1.2 API coordination, integration communication
Shace tariq	23I-3069	Backend Developer	2.1 Data Layer, 2.2 API Development, 2.3 Processing Logic
Hamad Khan	23I-3095	Frontend Developer	3.1 Dashboard Development, 3.2 Visualization, 3.3 Feature Integration
Ali Sajjad	23I-0110	QA & Testing Engineer	4.1 Unit Testing, 4.2 Integration Testing, 4.3 System Testing, 5.2 Integration

5.2 Resource Requirement Calendar

Role	12–27 Mar	28 Mar – 12 Apr	13–18 Apr	19–30 Apr	Hrs/day
Team Lead	Planning	Backend coordination	Integration	Final delivery	2-4
Backend Developer	Documentation	Backend development	Support	Integration	2-4
Frontend Developer	Documentation	UI development	Dashboard	Final UI fixes	2-4
QA Engineer	Documentation	Testing preparation	Testing	Final validation	2-4

6. Schedule Management Plan

Schedule Development

- The schedule baseline was developed from the WBS in Section 2.3 with duration estimates based on task complexity.
- ClickUp is used to track individual tasks, with labels for WBS phase and priority — consistent with the charter's communication plan.
- Critical path: MongoDB Schema Design → Event Ingestion API → Aggregation Service → Query API → React Dashboard → System Integration Testing.

Schedule Control

- Schedule is reviewed weekly on ClickUp.
- Tasks delayed by more than 2 days without recovery are escalated to the Integration Group on ClickUp.
- Milestone date shifts of more than 3 days require formal notification to the Integration Group via ClickUp team leads channel.
- The charter deadline of 'end of April' is a hard constraint for final integration and submission.

7. Quality Management Plan

Quality standards ensure the Analytics module delivers accurate insights and integrates cleanly with the Bring MERN system.

Quality Attribute	Standard / Metric	Measurement Method
Data Accuracy	Analytics counts within 5% of actual event counts	Compare ingested MongoDB events vs. aggregates
API Response Time	Express endpoints respond in < 2 seconds	Postman performance tests
API Reliability	All agreed endpoints return correct HTTP status codes	Integration test suite
Code Quality	No hardcoded secrets; modular Express routes; GitHub README	Integration Group review
Dashboard Usability	All React charts render in Chrome and Firefox	Manual browser testing
Documentation Completeness	All Assignment 2 required sections complete	Instructor evaluation rubric
Budget Compliance	Zero paid tools or services used (charter constraint)	Stack audit before deployment

8. Risk Management Plan

Risks are derived from the Project Charter risk section and module-specific concerns.

8.1 Risk Log

#	Risk	Prob.	Impact	Mitigation (per Charter)	Contingency
R 1	Integration delays – other modules miss event publishing deadlines, starving analytics of data	Medium	High	Early coordination via ClickUp; share event schema by 31 Mar	Use mock event generator for testing
R 2	Technology conflicts – modules use incompatible data formats or non-MERN tech	Low	High	Instructor consultation as per charter; Integration Group enforces MERN compliance	Implement adapter/transform layer
R 3	API contract changes post-freeze causing rework of ingestion or dashboard	Medium	High	Freeze API contracts at 31 Mar; all changes require Integration Group approval	Build adapter layer in ingestion service
R 4	Free-tier service downtime during final demo (Railway, Render, Atlas)	Low	High	Test deployment 1 week before demo; charter requires zero paid fallback	Run locally as fallback

9. Cost Baseline

Per the Project Charter: Total Budget = \$0. No paid APIs, services, or premium tools are allowed. The sub-project is fully compliant with this constraint.

Item	Estimated Cost	Free Tool Used	Actual Cost
Backend Hosting	\$0	Railway / Render free tier	\$0
Database Hosting	\$0	MongoDB Atlas free 512MB tier	\$0
Frontend Hosting	\$0	Vercel / Netlify free tier	\$0
Domain & SSL	\$0	Auto SSL from Vercel/Netlify subdomain	\$0
APIs / External Services	\$0	No external paid APIs used	\$0
Development Tools	\$0	VS Code, GitHub, Postman, ClickUp free tiers	\$0
TOTAL	\$0	Zero-budget constraint met	\$0

10. Quality Baseline

Quality Item	Acceptable Level	Comments
Event ingestion accuracy	>= 95% of events correctly stored in MongoDB	Tested with 500 synthetic events
Express API response time	< 2 seconds for all endpoints	Measured on free-tier deployment
React dashboard rendering	All charts render in Chrome and Firefox	Manual cross-browser check
Unit test coverage	>= 60% of core aggregation and API logic	Jest / Mocha test suite
Documentation completeness	All Assignment 2 required SPMP sections filled	Per Assignment 2 rubric
Integration compliance	All API contracts met; no breaking changes post-freeze	Verified by Integration Group
Budget compliance	\$0 spent – no paid tools	Per Project Charter constraint

11. Appendices

Appendix A: All Module Teams and Leads (from Project Charter)

Team Lead	Contact	Module
Rabeeha Kamran	i233025@isb.nu.edu.pk	User Authentication & Profile Management
Areej Hamid	i233041@isb.nu.edu.pk	Identity & Neighborhood Verification
Rana Ahmad	I233048@isb.nu.edu.pk	Neighborhood Creation & Management
Maryam Iftikhar	i233046@isb.nu.edu.pk	Community Posts & Feed
Muhammad Abdullah	I233010@isb.nu.edu.pk	Events & Activities Management
Muhammad Umer	(per charter)	Local Services Directory & Service Request & Matching
Abdul Haseeb	i233077@isb.nu.edu.pk	Community Marketplace (Buy/Sell)
Amna Malik	I222715@nu.edu.pk	Messaging & Communication

Usman Baig	i221509@nu.edu.pk	Trust, Rating & Reputation System
Aqsa	(per charter)	Moderation & Reporting / Conflict Resolution
Shah Faisal	(per charter)	Phishing / Spam / Fraud Detection
Muhammad Hamza (23I-3032)	i233032@isb.nu.edu.pk	Analytics & Community Insights

Appendix B: Analytics Module API Contract (v1.0 – Draft)

The Analytics module exposes the following REST API endpoints. Contract freeze date: 31 March 2026. All changes after this date require Integration Group approval via ClickUp.

Method	Endpoint	Consumer	Description
POST	/api/events	All modules	Ingest an event from any Bring module (requires User ID, Neighborhood ID)
GET	/api/analytics/summary	Admin dashboard, Integration	Platform-wide engagement summary
GET	/api/analytics/neighborhood/:id	Admin dashboard	Neighborhood-level stats and health score by Neighborhood ID
GET	/api/analytics/incidents	Admin dashboard, Moderation module	Incident frequency and type breakdown
GET	/api/analytics/export	Admin dashboard	Export analytics data as CSV for a given date range

Appendix C: Event Schema (v1.0 – shared with all module teams via ClickUp)

Field	Type	Description
event_type	string	e.g. 'post_created', 'incident_reported', 'item_listed', 'user_registered', 'message_sent'
module_source	string	Originating module, e.g. 'community_posts', 'marketplace', 'auth'
neighborhood_id	string (UUID)	Shared Neighborhood ID (common identifier per Project Charter)
user_id	string (UUID)	Shared User ID of the triggering user (common identifier per Project Charter)
timestamp	ISO 8601 string	Event timestamp, e.g. '2026-04-05T14:32:00Z'
metadata	object (optional)	Module-specific extra data, e.g. { category: 'theft', severity: 'high' }

12. Authorization Signatures

This Sub-Project Management Plan has been prepared by the module developer and is submitted as part of Assignment 2. All team leads and the Integration Group have agreed on the overall scope and module responsibilities during the kickoff meeting per the Project Charter process.

PREPARED BY (Group Laed) Name: shah Faisal Roll No.: 23I-0058 Email: i230058@isb.nu.edu.pk Module: Analytics & Community Insights Signature: _____ Date: 27 March 2026	APPROVED BY (Integration Group Lead) Name: Hassan Qureshi Roll No.: 23I-3029 Email: i233029@isb.nu.edu.pk Role: Integration Manager Signature: _____ Date: _____
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Integration Manager Name: Ahmed Ali Email: i233054@isb.nu.edu.pk Signature: _____ Date: _____	Integration Manager Name: Muhammad Irfan Email: i233065@isb.nu.edu.pk Signature: _____ Date: _____
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